

Knowing Where to Look Is Not Knowing What Is Right: A Difference-Aware Transformer for Physical Commonsense QA

Group 69

Abstract

PIQA asks which of two near-identical solutions achieves an everyday physical goal. We ask whether a model trained from scratch, without external data, can learn *where* the two solutions differ, and whether that is enough. Our Difference-Aware Contrastive Transformer (DACT) makes the difference explicit, with difference-tag embeddings, cross-solution attention and difference-guided pooling, and adds a jointly trained lexical head. The answer is yes to the first question and no to the second. Attention controls show that DACT learns to put 78% of its pooling attention on the differing words (uniform attention would give 25%), and together its components beat a vanilla Transformer by about 4 points. But knowing where to look is not knowing what is right: DACT is the best from-scratch model on validation (61.3%, 3 seeds), ahead of a TF-IDF baseline by 2 points in two independent runs, while on test the two are statistically tied (61.9 vs. 61.2%, McNemar $p=0.62$) and solve largely different items; pretrained models, with 65–800 times more parameters and web-scale pretraining, are 8–13 points better. Wrong predictions attend to the differing words even more than correct ones: the bottleneck is physical knowledge, not focus. We also report a pre-registered negative result and the failures that shaped the design.

1 Introduction

Physical commonsense, knowing that *baby wipes* rather than *index cards* remove chalk dust, is easy for people and hard for machines. PIQA (Bisk et al., 2020) tests it with a goal and two candidate solutions, exactly one of which is physically sensible. The task matters for assistants that give practical instructions and for robots that must choose feasible actions. Its key difficulty is that the two solutions are usually near-duplicates: in our training split the median share of differing solution to-

kens is 15%, so the answer hinges on a few substituted words whose plausibility requires world knowledge.

Related work. On PIQA, pretrained models are far from people: Bisk et al. (2020) report 77.1% for RoBERTa-large against 94.9% for humans, and multi-format QA training improves pretrained models further (Khashabi et al., 2020). Our setting differs, because the main model must learn from scratch. The closest architectural ideas come from multiple-choice reading comprehension and inference. The Option Comparison Network (Ran et al., 2019) and DCMN+ (Zhang et al., 2020) compare candidate answers with each other instead of scoring them independently, and ESIM (Chen et al., 2017) and the decomposable attention model (Parikh et al., 2016) align two texts token by token and build comparison features. DACT adapts these ideas to two near-identical solutions: difference tags and cross-solution attention make the alignment explicit, and difference-guided pooling focuses the decision on the substituted words. Finally, work on annotation artefacts shows that shallow lexical cues can solve many items without reasoning (Gururangan et al., 2018; Poliak et al., 2018); our TF-IDF baseline and lexical head make this effect measurable on PIQA.

Our aim was a model that is trained from scratch but built around this structure. DACT (Section 2) makes the difference between the solutions explicit at every stage: in the input, in the comparison and in the summary vector, and it adds a lexical head that scores which words make a solution plausible. We compare it with five baselines, from majority vote to a zero-shot LLM, run nine ablations with three seeds each, and analyse the attention with controls and probes.

Our main findings are: (i) a from-scratch model does learn where the options differ: the attention focus on the differing words is learned, not built

in, and the difference-aware design as a whole clearly beats a vanilla Transformer; (ii) this is not enough: DACT is the best from-scratch model on validation but statistically tied with a strong lexical baseline on test; (iii) errors come from *judging* the differing words without physical knowledge, which only pretraining supplies. Section 5 documents what worked and what did not during development.

2 Methodology

Input and difference tags. For a goal g and solutions s_1, s_2 , each option k is encoded as $x^{(k)} = [\text{CLS}] g [\text{SEP}] s_k [\text{SEP}]$ with a BPE vocabulary (Sennrich et al., 2016) learned on the training split. We align the token sequences of s_1 and s_2 with difflib and give every solution token a tag $t_i \in \{\text{shared}, \text{different}\}$ (goal and special tokens: *other*). The input embedding is

$$e_i = E_{\text{tok}}[x_i] + E_{\text{pos}}[i] + E_{\text{seg}}[\sigma_i] + E_{\text{tag}}[t_i]. \quad (1)$$

The tag embedding (our contribution) tells the encoder from the first layer where the options disagree.

Encoder. A shared pre-LayerNorm Transformer encoder (Vaswani et al., 2017; Xiong et al., 2020) maps each option to hidden states $H^{(k)} \in \mathbb{R}^{L \times d}$.

Contrastive cross-solution attention. Each token of option k attends to the solution tokens of the rival option $\bar{k}$ plus its CLS token as a “no counterpart” sink:

$$c_i = \text{MHA}(h_i^{(k)}, H^{(\bar{k})}, H^{(\bar{k})}), \quad (2)$$

and the states are updated with the comparison features of ESIM (Chen et al., 2017): $\tilde{h}_i = \text{LN}(h_i + \text{FFN}([h_i; c_i; h_i - c_i; h_i \odot c_i]))$. Adapting ESIM’s local inference from premise–hypothesis pairs to two competing answers is our design choice: plausibility is relative, so each option is represented by what it has *instead of* its rival.

Difference-guided pooling. Over the solution tokens we use additive attention (Bahdanau et al., 2015) with a learned bias per tag (our extension):

$$\alpha_i \propto \exp(v^\top \tanh(W\tilde{h}_i) + b_{t_i}), \quad p = \sum_i \alpha_i \tilde{h}_i, \quad (3)$$

with $b_{\text{different}}$ initialised to 1 and the others to 0. A scorer MLP maps $[p; \tilde{h}_{\text{CLS}}]$ to a scalar s_k .

Lexical head. Following the wide & deep idea (Cheng et al., 2016), a sparse linear component scores the words of each solution directly: every BPE unigram and bigram of s_k is hashed into one of 2^{18} buckets with learned weight w_j (initialised to 0),

$$\ell_k = \sum_{j \in \text{ngrams}(s_k)} w_{h(j)}, \quad z_k = s_k + \ell_k. \quad (4)$$

It is trained jointly with the Transformer, with its own learning rate and weight decay. Under the pairwise softmax only $\ell_1 - \ell_2$ matters, so shared n-grams cancel and the head learns which substituted words indicate a plausible solution, the signal that a TF-IDF baseline exploits.

Objective. The two scores compete in a softmax, $P(k) = \text{softmax}(z_1, z_2)_k$, trained with cross-entropy and label smoothing 0.1, which matches the forced-choice task. Every component can be switched off in a single configuration object, which is how the ablations are run.

3 Experimental Setup

Data. PIQA provides 16,113 training items and 1,838 labelled development items; the latter serve as our test set. Test labels are used only in the final evaluation step (enforced by a guard in the notebook), although the test file is loaded and tokenised at the start and B1/B3 compute (but do not inspect) their test predictions earlier. An audit of the data found six duplicate rows within the training file and one test item (index 1545) that appears twice in our training split. We keep the split unchanged and disclose the overlap: re-scoring the saved test predictions without that item changes accuracy by about 0.02 points (DACT $61.86 \rightarrow 61.84\%$, TF-IDF $61.21 \rightarrow 61.19\%$). We hold out a stratified 10% of the training file (seed 42) for validation: 14,501 / 1,612 / 1,838 items, all label-balanced (50.0 / 50.0 / 50.5% label 1). Text is whitespace-normalised, NFKC-normalised and lower-cased. Goals are truncated to 40 and solutions to 96 BPE tokens; for longer solutions (1.6% of test solutions) we keep the 96-token window centred on the differing span, not the prefix (Section 5).

Training. AdamW (Loshchilov and Hutter, 2019) ($\beta_2=0.98$, weight decay 0.05, learning rate 3×10^{-4} ; lexical head 10^{-2}), 6% linear warm-up then cosine decay, batch 128, gradient clipping 1.0, dropout 0.2, at most 20 epochs with early

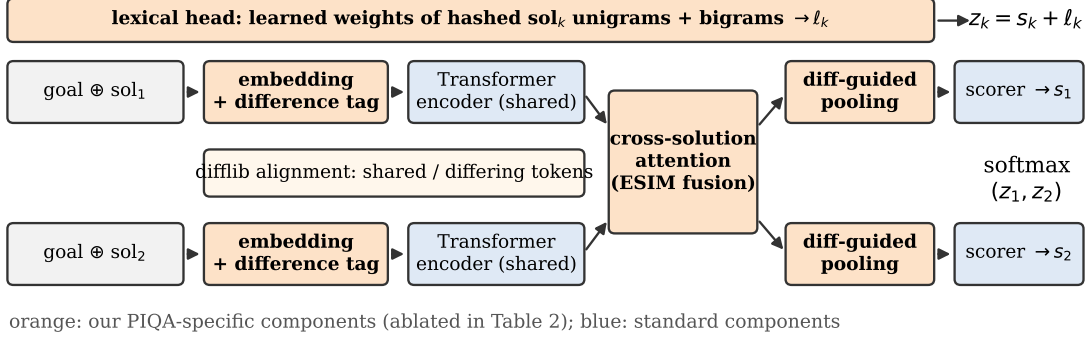


Figure 1: DACT. Both options pass through the same weights, so for already-encoded inputs swapping the options swaps the scores (verified in the notebook). The full pipeline is not exactly equivariant, because the difflib alignment depends on option order (Section 5).

stopping on validation accuracy (patience 6), fp16 mixed precision on a free Colab T4. Step 1 of the notebook selects the configuration on validation with one seed from five candidates: base ($d=256$, 4 layers) 61.8%, small ($d=128$, 2 layers, 4 heads) 61.4%, base with dropout 0.3 60.8%, and base/small with a 10-epoch in-domain masked-LM warm-up on the training split (Devlin et al., 2019) 60.8 / **61.8%**. The selected model is **small with the warm-up**, 0.06 points ahead of base; it has 1.93 M parameters, 262 k of them in the lexical head. The three best candidates lie within 0.43 points, less than one seed std of the final model (0.5), and an earlier A100 run of the same code selected small without the warm-up. We therefore read the grid as ruling out clearly worse settings (larger dropout, warm-up on the base model), not as ranking the top candidates; Section 5 discusses this seed sensitivity. The lexical head’s learning rate was chosen in a separate validation pilot.

Baselines. All baseline numbers come from our own runs. **B0** majority class. **B1** TF-IDF of 1–2-grams of $s_2 - s_1$ with logistic regression, C tuned on validation; it tests how far shallow lexical cues go (Gururangan et al., 2018). **B2** a 2-layer BiLSTM with additive attention and the same tokenizer (5.1 M parameters), a recurrent model of similar budget. **B3** roberta-base (Liu et al., 2019) (125 M parameters) fine-tuned with a multiple-choice head (4 epochs, learning rate 2×10^{-5} , 3 seeds, best epoch on validation), representing pretrained encoders; a seed below 52% validation accuracy after epoch 1 is restarted with a new seed (declared before the run). Because fine-

tuning proved unstable, we also re-ran B3 with unchanged settings on validation only, as part of a pre-registered stability test. **B4** Qwen2.5-1.5B (Qwen Team, 2024) zero-shot: the prompt “Goal: g / Solution:” followed by each solution, choosing the higher length-normalised log-likelihood, which measures LLM knowledge without any training.

Metrics. Accuracy is PIQA’s official metric; labels are balanced and each item has exactly one correct option, so it is unbiased with 50% chance level. Trained models are run with seeds 42–44 and reported as mean $\pm$ std. For the seed with the best *validation* accuracy we give a 95% bootstrap CI (2,000 resamples; Efron and Tibshirani, 1993) and an exact McNemar test (McNemar, 1947) against DACT on the test set.

Validity of the test numbers. Validation accuracy is the only number that reflects design decisions. The test set was evaluated in several development runs, and the lexical head was introduced after earlier test results had been seen (Section 5). Test numbers are therefore reported for completeness and comparison, not as a clean held-out estimate of generalisation.

4 Results and Analysis

Overall comparison. DACT is the best from-scratch model on validation and on test (Table 1): +2.0 / +3.1 points over the BiLSTM (McNemar $p=0.09$ on test) and +2.0 / +0.7 over TF-IDF. Against TF-IDF the picture is split by design: DACT’s 2-point validation lead appeared in both of our independent runs (A100 and T4), whereas on test the two are statistically tied ($p=0.62$; $p=0.88$

Model	Val	Test
<i>trained from scratch</i>		
B0 majority	50.0	50.5
B1 TF-IDF + LR	59.2	61.2
B2 BiLSTM + attention	59.3 ± 0.7	58.7 ± 1.1
DACT (ours)	61.3 ± 0.5	61.9 ± 0.3
<i>pretrained</i>		
B3 RoBERTa-base, fine-tuned [†]	59.2 ± 5.8	57.3 ± 6.9
validation re-run [‡]	69.0 ± 0.6	–
B4 Qwen2.5-1.5B, zero-shot	78.2	75.1

Table 1: Accuracy (%), mean $\pm$ std over 3 seeds; B0, B1 and B4 are deterministic single runs. DACT test 95% CI: 59.6–64.0. [†]Final run: two of three seeds did not leave chance-level training loss (ln 2); the working seed reaches 65.3% (test CI 62.4–66.6). [‡]Same code, settings and GPU type, all three seeds trained (69.6 / 68.5 / 68.8); validation only, so no test number exists. An A100 run gave 68.0 / 66.7.

Ablation (vs. full DACT)	Δ Val	Δ Test
– lexical head	–2.3	–0.3
pointwise (independent BCE) objective	–1.7	–1.4
– difference tags	–0.2	–1.3
mean instead of attention pooling	–0.2	+0.2
– MLM warm-up	+0.1	–0.3
– tag bias in pooling	+0.1	–0.6
tag bias initialised at 0	+0.2	–0.7
– cross-solution attention	+0.6	–0.6
vanilla (four components removed)	–3.7	–4.3

Table 2: Ablations, 3 seeds each, one change at a time on the selected configuration; differences of the unrounded 3-seed mean accuracies, rounded to 0.1 points (seed std $\approx$ 0.1–1.1).

in the A100 run). A tie with a bag-of-words model is not a null result here: DACT reaches it with 1.93 M parameters trained on 14.5 k items, and the two models make substantially different errors (below), so the tie reflects two different sources of evidence of similar strength, not one model reproducing the other. The gap to pretraining is the decisive one: the zero-shot LLM (about 800 times more parameters) is 13.2 points above DACT on test, and RoBERTa (65 times more parameters), when its fine-tuning succeeds, is about 8 points above on validation (69.0 vs. 61.3%). Fine-tuning success is itself a random variable: with identical settings, one T4 session trained one of three seeds and another trained all three, a known instability (Dodge et al., 2020) that the mean of a single run hides.

Ablations. Each ablation tests one hypothesis about a design choice (Table 2); everything else, including seeds, data and budget, is fixed. *The de-*

sign helps as a whole: removing the tags, cross-solution attention, attention pooling and the lexical head together costs 3.7 / 4.3 points, the largest drop and the only significant one on test ($p=0.007$). *Direct lexical evidence helps*, mainly on validation: removing the lexical head costs 2.3 / 0.3. The forced-choice objective helps on both splits (independent binary scoring: 1.7 / 1.4). The difference tags help on test only (0.2 / 1.3). Single structural components make no consistent difference: cross-solution attention (removing it gives +0.6 / –0.6), attention pooling (+0.2 on test with mean pooling), the tag bias and its initial value, and the MLM warm-up (+0.1 / –0.3) all move by less than a point in opposite directions on the two splits. They overlap: the tags already tell the model where the options differ, so each one is replaceable by the others. Differences below one seed std are not conclusive, and with nine comparisons single p-values should not be over-read (Dror et al., 2018).

Where the design succeeds. Two results separate DACT’s structure from its lexical component. First, removing the difference-aware components together (the vanilla ablation) costs 3.7 / 4.3 points, the only significant ablation on test ($p=0.007$), and the same comparison cost 2.5 / 2.4 in the A100 run: the combined design helps on both splits in both runs. Second, DACT and TF-IDF agree on only 78.2% of test items; DACT alone solves 11.2% (206 items) and TF-IDF alone 10.6%, for an oracle accuracy of 72.4%. The items only DACT solves include minimal pairs with a one-word difference inside long shared text (*release* vs. *do not release* the ignition key; the typo *sparrow bottle* vs. *spray bottle*), where TF-IDF’s difference vector $s_2 - s_1$ is close to zero and the alignment-based tags are most informative. Against RoBERTa the oracle rises to 82.4%, again pointing to complementary evidence rather than redundancy.

Is the attention focus learned? The trained model puts 78.0% of its pooling attention on differing tokens, which make up 24.8% of solution tokens (Figure 2). Because the tag bias starts at 1, we separate the sources: an untrained model reaches only 38.9%, while the trained model with the bias set to 0 still reaches 62.0%, and the bias itself barely moved in training (1.0 $\rightarrow$ 1.03). The focus is therefore learned by the pooling scorer, which also explains why initialising the bias at 0

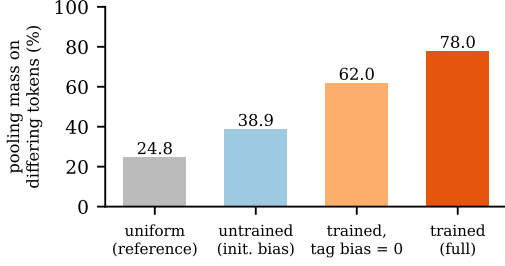


Figure 2: Share of pooling attention on differing tokens (test set). Removing the tag bias from the trained model keeps most of the focus, so it is learned, not built in.

changes nothing. Attention weights are not a complete explanation of a decision (Jain and Wallace, 2019; Wiegrefe and Pinter, 2019), so we use them as evidence of where the model looks, not of why it decides.

Why does the neural part alone not beat bag-of-words? We tested two explanations on validation, which also explain the test tie above. *H1: the pooling does not select* is rejected: the normalised pooling entropy is 0.59 (1 = uniform), for correct and wrong predictions alike ($p=0.98$). *H2: the difference signal is missing*: a linear probe separating differing from shared tokens reaches 62.0% balanced accuracy on the vanilla Transformer’s embeddings and only 63.0% after its encoder, so without tags the encoder barely works out where the options differ, whereas in DACT the tag remains perfectly decodable (100%) in every layer. Tags and the lexical head thus supply information the encoder does not learn from 14.5 k items. Crucially, wrong predictions put *more* attention on the differing tokens than correct ones (79.0 vs. 77.4%, $p=0.03$): looking at the right words is not the bottleneck; judging them is.

Qualitative analysis. For the item *adding scents to lotion bars* (Figure 3, top), the wrong option adds the oils “to the water in the bottom of the double boiler”. The pooling peaks on *bottom* (of the double boiler), the part that makes the option wrong, yet the model prefers it with $p = 0.96$: it locates the decisive span but cannot know that oil in the water bath never reaches the lotion. In *a small but cheap home* (bottom), the options differ in a one-letter typo, a *shipping* vs. a *shop-ping* container, plus *living space* vs. *relaxing environment*. Pooling puts almost half its weight on *shipping* and selects the differing spans on both sides, but the scores tie: the model sees the dif-

Step	Change	Val / Test	Outcome
Run 1 (A100)	original design, prefix truncation	60.1 / 60.4	below TF-IDF on test
Run 2 (T4)	difference-aware truncation, attention controls	59.4 / 60.2	focus is learned
Pilot (val)	lexical head: shared $\rightarrow$ own learning rate	59.5 $\rightarrow$ 60.9	own rate needed
Run 3 (T4)	lexical head	61.1 / 61.6	above TF-IDF on val
Run 4 (A100)	lexical head, final code base	61.2 / 61.6	reproduces run 3
Pre-reg. (val)	goal-matching block	60.9 vs. 61.3	rejected, no test run
Final (T4)	submitted code	61.3 / 61.9	reproduces run 4

Table 3: Development history: DACT accuracy (%). TF-IDF + LR scores 59.2 / 61.2 in every run.

ference but does not know that shipping containers are a common cheap home. Successes rest on the same word-level evidence: for *a simple camping lantern*, the gold option (a milk jug on top of an *LED light, translucent plastic*) wins with $p = 0.99$ against *fairy lights* in a filled jug, with pooling spread over the many differing words. Finally, the most confident error, *to make the most of an exotic trip* ($p = 0.98$ for a strict itinerary over keeping plans flexible), shows the lexical head’s calibration problem: the two options share almost no words, and since ℓ_k sums over n-grams, long differing spans receive extreme scores.

5 What Worked and What Did Not

Table 3 summarises how the final model came about. All runs are logged in the repository; only the final T4 run of the submitted code is used in Tables 1–2.

What worked. Difference-aware truncation removed inputs that had become identical after prefix truncation (in run 1, one item differed only at BPE positions 97–98). The attention controls (Figure 2) turned the worry that the focus was built in by the bias initialisation into a test, which it passed. Three RoBERTa seeds exposed fine-tuning instability: in run 2 one seed stayed at chance (53.7% validation), a known failure (Dodge et al., 2020); in the final run the declared restart rule replaced one such seed, but two more stalled just above its threshold. Giving the lexical head its own learning rate made it useful.

Goal: Adding scents to lotion bars

Option 1, $p = 0.96$ grate the wax and melt, in a double boiler, add the oils to the water in the bottom of the double boiler.

Option 2 (gold), $p = 0.04$ once the lotion and wax have melted, and reached the trace stage, add the scented oils and stir to combine.

Goal: How to make a small but cheap home easily?

Option 1 (gold), $p = 0.50$ purchase a shipping container online and have it installed on a cheap rental lot, then convert the container into a living space

Option 2, $p = 0.50$ purchase a shopping container online and have it installed on a cheap rental lot, then convert the container into a relaxing environment

Figure 3: Pooling attention over each solution (darker = higher weight; **bold** = tokens that differ between the two options). Top: *adding scents to lotion bars*, a confident error ($p = 0.96$ for option 1). Bottom: *a small but cheap home*, the most uncertain item ($p = 0.50$; option 1 is gold).

What did not. The first lexical head barely moved from zero before early stopping. The goal-matching block (165 k parameters) was pre-registered, with its specification, acceptance rule (validation gain ≥ 1.0 point and > 2 pooled seed std) and decision procedure committed before training (experiments/goal_matching/); it lost 0.3 points on validation and was rejected without a test run. On the free T4, emulated bf16 slowed training from 108 to 377 ms per step until bf16 was restricted to GPUs that support it natively. Seed sensitivity was the main methodological lesson. Between run 4 and the final run, single ablations and the from-scratch baselines moved by up to 1.6 points, the one-seed grid chose a different configuration, and RoBERTa’s fine-tuning succeeded in one, two or three of three seeds depending on the session. With a free-GPU budget (three seeds per model, one per grid candidate), our conclusions rest on effects larger than this spread (the vanilla ablation, the pretraining gap).

Protocol, audit and evidence. An independent audit of run 1 (docs/PROJECT_AUDIT.md) found the training/test overlap (Section 3), the order-dependent alignment (103 test items in the original pipeline, 102 now; an opt-in canonical alignment removes it), early loading of the test file and unsaved checkpoints, and added prediction export, provenance and validation checks. The submitted notebook contains the complete final run (commit b880108; 57 cells, no errors, logs of all 33 training runs) and a cell that checks tables against the result files; test labels are read only in its final step, and every read is logged; docs/DEVELOPMENT_LOG.md

records every step, including the failures.

6 Conclusion

DACT shows that task-specific structure pays off when training from scratch: making the difference between two near-identical solutions explicit and adding direct lexical evidence yield the best from-scratch model on validation, consistently across two independent runs, and a model that ties a strong lexical baseline on test while solving a different 11% of the items. The combined difference-aware design is the robust contribution (3.7 / 4.3 points over a vanilla Transformer); single structural components are interchangeable, because they carry overlapping information about where the options differ. Its attention demonstrably learns to focus on the decisive words. The analysis equally shows the limit of the approach: the model finds *where* the options differ but judges *which* is right with word statistics from 14.5 k items, while models with 65–800 times more parameters and pretraining reach 69–75%.

Limitations. Besides the seed sensitivity, test-protocol and option-order caveats above: the lexical head is over-confident on long differing spans; the model overfits quickly (85–92% training against about 61% validation accuracy at the selected epoch), so early stopping does most of the regularisation; and ablations that remove modules also change parameter counts. Next steps: length-normalised lexical scores and the difference-aware design on pretrained representations.

Team Contributions

The work was divided among the three members; each implemented and tested his own part of the code and wrote the report sections that describe it.

Bayron: original DACT pipeline (data processing, model, training and evaluation code, baselines B0–B4), the original A100 run, the final A100 runs of both versions and their comparison, and integration of the final notebook and report.

Ali: model revision (difference-aware truncation, lexical head and its validation pilot), attention controls and diagnostics, three-seed RoBERTa baseline, the free-T4 runs, and the first draft of the report.

Salah: evidence audit of the original run (training/test overlap, order-dependent alignment, test exposure), robustness and provenance code (validation, determinism, prediction export), and the local replication of the original model.

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